PRIME FACTOR STUDIO — IN-BOX USER GUIDE

Kinesthetic Number Theory, Proportional Logarithmic Stacking & 12-Socket Lid Tray

SERIES 1 • CARD 05 / 12 • DOMAIN: FUNDAMENTAL THEOREM OF ARITHMETIC

CARD 05 / 12 • PRIME-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

FEEL THE PRIMES. MULTIPLY WITH HEIGHT.

Standard base-10 counting blocks add lengths (1+1=2).

Prime Factor blocks are logarithmically proportional:

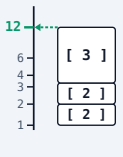
log(a × b) = log(a) + log(b)

Stacking blocks in a tower multiplies their numbers!

CALIBRATED LOGARITHMIC DATUM

- Unit Datum: Block [2] = 1U = 7.00 mm visible height
- Log Modulus: K_log = 23.2535 mm / decade
- Prime Formula: h_vis(p) = K_log × log10(p)

VISUAL PROOF: TOWER 12 = [2] + [2] + [3]



▲ SUMMIT = 12

2 × 2 × 3 = 12

Visible: 25.09 mm

Commutative:

[3]+[2]+[2] = same height!

2. KIT PARTS INVENTORY (23 PIECES)

COMPONENT	QTY	SPEC	COLOR
Prime Block [2]	2x	7.00 mm	Matte Blue
Prime Block [2²]	2x	14.00 mm	Matte Blue
Prime Block [2³]	2x	21.00 mm	Matte Blue
Prime Block [3]	4x	11.09 mm	Muted Red
Prime Block [3²]	1x	22.18 mm	Muted Red
Prime Block [5]	2x	16.25 mm	PLA White
Prime Block [7]	2x	19.65 mm	PLA Gray
Prime Block [11]	1x	24.21 mm	PLA Tan
Prime Block [13]	1x	25.90 mm	PLA Tan
Prime Block [17]	1x	28.61 mm	PLA Gray
Prime Block [19]	1x	29.73 mm	PLA Gray
Prime Block [23]	1x	31.67 mm	PLA White
Prime Block [29]	1x	33.51 mm	PLA White

TOTAL: 21 Blocks + 7U Box + Stage Lid = 23 Parts (7U Farm) • 47 Parts (13U)

SLANT 3D PRODUCTION PALETTE

- PLA Matte Blue: Prime [2] family (2, 2², 2³).
- PLA Matte Muted Red: Prime [3] family (3, 3²).
- PLA White: Prime [5], Primes [23], [29], Enclosure Box & Lid.
- PLA Tan: Mid primes [11], [13] • PLA Gray: Primes [7], [17], [19].

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Stand prime blocks in product-labeled sockets (2^{12}=4096, 3^7·2=4374, etc). All 12 towers level off at the 84mm waterline. Snap stage lid flush for silent carry.

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts
Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!

Keep stored inside box lid!

FUNDAMENTAL THEOREM

Every integer > 1 has a unique prime factorization decomposition.

STACKCALC ARCHIVAL INSTRUMENT LABS
Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL
stackcalc.io/parts

LESSON GUIDE: HOW TO FIND THE LEAST COMMON MULTIPLE (LCM)

Step-by-Step Hands-On Tutorial: Finding LCM(12, 18) = 36 & The Law of Conservation of Primes

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LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/prime-studio/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS 6.NS.B.4 (LCM & GCD)
- CCSS 4.OA.B.4 (Factor Pairs)
- TEKS 111.26.B.6.7.A (Prime Factor...)

Grades 4–9 Math & Pre-Algebra

GUIDING INQUIRY

Which prime blocks are shared by both towers, and which blocks are unique to one tower?

1 STEP 1: FACTOR & BUILD TOWERS 12 & 18

Decompose numbers 12 and 18 into prime building blocks:

- Tower 12: Stack [2] + [2] + [3] (2 × 2 × 3 = 12)
- Tower 18: Stack [2] + [3] + [3] (2 × 3 × 3 = 18)

2 STEP 2: FIND SHARED OVERLAP (THE GCD)

Stand Tower 12 and Tower 18 side-by-side. Identify the blocks they share in common (intersection):

- Tower 12 has: [2], [2], [3] | Tower 18 has: [2], [3], [3]

Shared Overlap: Exactly one [2] and one

3–4 STEPS 3 & 4: ASSEMBLE THE LCM TOWER

3. TAKE PRIME UNION: Take highest power of each prime: two [2]s, two [3]s.

4. STACK LCM TOWER: [2] + [2] + [3] + [3] = 36 (Summit: 36.19 mm)!

Calculator Verification: Press Factor Tree Icon ([PRIME] / [g][FACT]) and run 12 [ENTER] 18 [GCF] 12 [ENTER] 18 [f] [LCM] → 36.

5 CORE PRINCIPLE: CONSERVATION OF PRIMES: A × B = GCD × LCM

Multiply Tower 12 and Tower 18: you have three [2]s and three [3]s. Look at your GCD Tower ([2], [3]) and your LCM Tower ([2]², [3]²): together they contain the EXACT same prime blocks!

Therefore: A × B = GCD(A, B) × LCM(A, B) = 6 × 36 = 216!

★ EXTENSION INVESTIGATION: FIND LCM(8, 12)

Build Tower 8: [2]+[2]+[2] and Tower 12: [2]+[2]+[3]. Prime union takes three [2]s and one [3] → LCM = 24 (28.09mm)! (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.